

# Bike Data Analysis

SQL-driven insights into bike-sharing usage patterns, revenue generation, and profitability



# Project Overview



## Data Integration

Combined yearly bike rental datasets using SQL Server Management Studio



## Financial Metrics

Calculated revenue and profit per ride with pricing and cost data



## Pattern Analysis

Examined seasonal, weekday, and rider type distribution patterns



# Three Datasets

## Bike Share Year 0

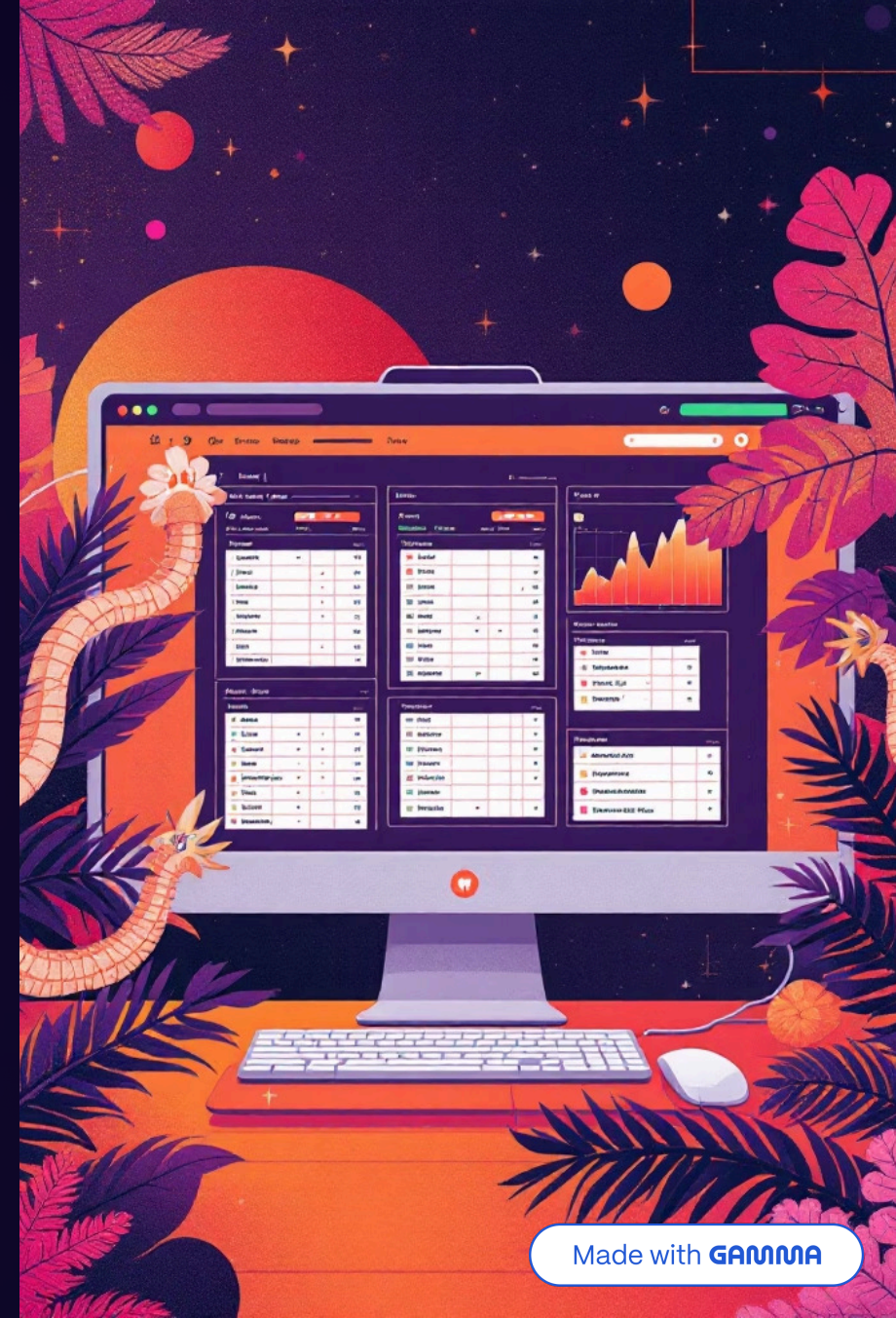
First year of rental data with timestamps, seasons, weekdays, hours, rider types, and rental counts

## Bike Share Year 1

Second year with identical structure for comparison and trend analysis

## Cost Table

Pricing per ride and cost of goods sold for financial calculations



# Data Preparation



## CTE Creation

Common Table Expression to structure queries



## Union All

Merged records from both years into single dataset



## Left Join

Attached pricing and cost information to each record

# Key Metrics Generated

2

## Years Combined

Complete dataset  
integration

11

## Core Metrics

Riders, revenue, profit,  
patterns

3

## Dataset Types

Share data and cost  
table



# Calculated Metrics

## Revenue Formula

$$\text{Revenue} = \text{Riders} \times \text{Price}$$

Revenue generated per ride based on rider count and pricing

## Profit Formula

$$\text{Profit} = \text{Revenue} - \text{COGS}$$

Profit per ride after subtracting cost of goods sold



# Business Insights



## Seasonal Trends

Rental variations across seasons driven by weather conditions and rider preferences



## Peak Hours

Hourly analysis reveals highest rental periods during commuting times



## Rider Types

Different customer segments contribute uniquely to total usage patterns

# Tools & Techniques

1

## SQL Server Management Studio

Primary platform for data analysis and query execution

2

## SQL Queries

Data aggregation, joins, and transformations using structured queries

3

## Common Table Expressions

CTE for organizing complex multi-step queries efficiently





# Project Outcome

01

## Dataset Combination

Successfully merged multi-year data into unified analysis framework

02

## Financial Insights

Generated revenue and profit metrics for business evaluation

03

## Operational Optimization

Provided actionable insights for bike-sharing service improvements



# Strategic Recommendations

## Conservative Pricing

10-15% increase to test market response without hitting price ceiling

## Segmented Strategy

Different pricing for casual vs. registered users based on sensitivity

## Monitor & Adjust

Implement new prices with close monitoring of customer feedback and sales data